

TECHNICAL REVIEW

multipgs

Theory, methods, evidence, and documentation

Reviewed state	Package version 0.3.0, commit 0ce9ad8888356dc5834f5e8c323621765ac7afb3
Review date	12 August 2026
Scope	Mathematical theory, three fitting routes, implementation contracts, checked-in benchmarks, user documentation, tests, and distribution packaging
Test baseline	294 tests passed on Python 3.12.13, NumPy 2.5.2, scikit-learn 1.9.0, pytest 9.1.1, and local ldpred3 0.4.6
Artifact status	This report is descriptive and diagnostic. It is not a clinical validation, regulatory assessment, or substitute for independent outcome evaluation.

Abstract

multips combines multiple polygenic scores through three distinct information routes: individual-level penalised stacking, score-space summary-statistic stacking, and a training-free same-trait meta-score. The selection-index theory is sound, the individual and summary Gaussian objectives are algebraically coherent when their score scales are handled correctly, and the repository contains unusually extensive benchmark provenance. The review also found one high-severity correctness defect in independent-GWAS tuning, one unsafe user-facing recommendation for decorrelated weights, and several documentation and provenance inconsistencies. Existing simulations support estimator sanity and moment identities, but they do not establish clinical utility, ancestry portability, or real-world comparative superiority.

Executive conclusions

Overall assessment. The mathematical core is credible. The reviewed state has broad documentation, 294 passing tests, and 34 checked benchmark artifacts. Its evidence should nevertheless be described as a combination of algebraic checks, controlled simulations, and real-data diagnostics - not as end-to-end clinical validation.

Route-blocking finding. Independent summary-statistic tuning forms a non-orthogonal basis and applies BB^T as if it were an orthogonal projection. A full-rank two-score example discards 95.6% of both tuning and training moments when the mathematically correct discarded fraction is zero.

Documentation finding. The README and guide recommend decorrelated with Daetwyler-derived accuracy, while the checked real CAD diagnostic gives R^2 0.1773 without decorrelation and 0.00226 with it. The rule is theoretically optimal only when its per-score target correlations are accurately specified and consistently oriented.

Table 1: Decision summary for the reviewed package state.

Area	Conclusion	Recommended disposition
Theory	Selection-index optimum, two-score gain, Daetwyler scaling, and score-space moments are internally coherent under stated assumptions.	Retain; tighten assumptions and notation.
Individual fit	Strong simulation sanity checks; CMSA and the nested fallback are package-specific departures from the published Multi-PGS workflow.	Usable with independent external assessment.
Summary fit	Exact-moment simulations agree closely with individual fitting, but independent tuning contains a confirmed projection defect.	Block independent tuning until fixed and regression-tested.
Training-free fit	Simple rules are transparent; decorrelation can amplify small accuracy misspecification catastrophically.	Default to the undecorrelated rule unless component accuracy is independently credible.
Evidence	294 tests pass and 34 benchmark artifacts are checked in; real outcome validation and ancestry-transfer evidence are absent.	Describe evidence level precisely; do not imply clinical validity.
Documentation	Broad and candid, but several high-impact statements have drifted from code and checked results.	Correct the user-facing recipe and automate result-to-prose checks.

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1 Scope and review method

The review covered the public modules under `multipgs/`, the README, theory, algorithm, guide, API map, bibliography, tests, build metadata, and all checked benchmark summaries and provenance records. The report is pinned to commit `0ce9ad8`; later source or benchmark changes can invalidate its line-level findings.

The evidence was classified into four levels:

1. algebraic identities and direct numerical counterexamples;
2. unit and contract tests;
3. replicated synthetic or real-LD simulations with checked raw rows;
4. descriptive real-data diagnostics without an untouched outcome truth.

These levels are not pooled. A passing simulation establishes behaviour under that simulation, not biological transport or clinical benefit.

The published Multi-PGS paper trained standardised PGS with covariates and reported fivefold out-of-sample assessment [9]. `multipgs` implements the same broad estimator, but substitutes CMSA, adds a nested fallback gate, supports a package-specific summary-statistic objective, and uses a different incremental- R^2 convention. Those differences are features, not clerical details, and are made explicit below.

2 Statistical theory

2.1 Notation and score scale

Raw score S_{ik} is the allele-weighted score for individual i and component k . Penalisation is defined after training-data standardisation:

$$Z_{ik} = \frac{S_{ik} - \mu_k}{\sigma_k}, \quad \beta_k = \frac{\theta_k}{\sigma_k}, \quad (1)$$

where θ_k is the standardised-score coefficient and β_k is the raw-score coefficient returned for deployment. Confusing these coordinates changes the meaning of every penalty factor.

Table 2: Notation used consistently in this report.

Symbol	Meaning
n, q	number of individuals and number of component scores
S, Z	raw and training-standardised score matrices
θ, β	standardised-score and raw-score combination coefficients, related by Equation (1)
g_k, g_f	standardised additive genetic values for component trait k and focal trait f
R_k	$\text{cor}(Z_k, g_k)$, genetic-value accuracy of score k for its own trait
$r_G(k, \ell)$	target-population genetic correlation between traits k and ℓ
ρ	target-population vector with $\rho_k = \text{cor}(Z_k, g_f)$
C	correlation matrix of the standardised component scores in the target population
N_k, M	effective GWAS sample size and effective number of independent effects or segments
π, p_s	population prevalence and sample case fraction for a binary trait; these symbols avoid reusing the score count q

2.2 Selection-index model

A useful working model writes each standardised component score as

$$Z_k = R_k g_k + \sqrt{1 - R_k^2} \varepsilon_k, \quad (2)$$

where $\text{var}(\varepsilon_k) = 1$, every ε_k is orthogonal to every genetic value, and the residual errors may be correlated across scores. Under independent residual errors,

$$\rho_k = R_k r_G(k, f), \quad C_{k\ell} = \begin{cases} 1, & k = \ell, \\ R_k R_\ell r_G(k, \ell), & k \neq \ell. \end{cases} \quad (3)$$

Discovery sample overlap is one mechanism that can correlate the residual errors; shared pipelines, LD references, model fitting, and other common estimation artifacts can do so as well. Conversely, overlap need not induce a large covariance in every design. The word *exactly* is therefore unwarranted.

For a weight vector w , the squared genetic-value correlation of the linear index is

$$R^2(w) = \frac{(w^\top \rho)^2}{w^\top C w}. \quad (4)$$

When C is positive definite, Cauchy-Schwarz gives

$$w_\star \propto C^{-1} \rho, \quad R_{\max}^2 = \rho^\top C^{-1} \rho. \quad (5)$$

This is the classical selection-index solution [1, 2]. A pseudoinverse or ridge is needed when the score correlation matrix is singular, but neither can repair a misspecified ρ .

2.3 Closed-form gain from one auxiliary trait

For the focal score and one auxiliary score with genetic correlation r , the exact increment above the focal score is

$$\Delta R^2 = \frac{r^2 R_k^2 (1 - R_f^2)^2}{1 - r^2 R_f^2 R_k^2}. \quad (6)$$

The three main drivers are visible: squared genetic correlation, squared auxiliary accuracy, and the squared information still missing from the focal score. A large auxiliary GWAS cannot compensate for genetic irrelevance, and a well-powered focal score leaves less room to gain.

The Daetwyler reference model relates genetic-value accuracy to GWAS power:

$$R_k^2 = \frac{x_k}{1 + x_k}, \quad x_k = \frac{N_k h_k^2}{M}. \quad (7)$$

It is an upper-reference model under strong architecture and population assumptions, not a per-score truth oracle [3]. The package function `daetwyler_r2` returns phenotypic R^2 , $h_k^2 R_k^2$, so taking its square root gives $h_k R_k$.

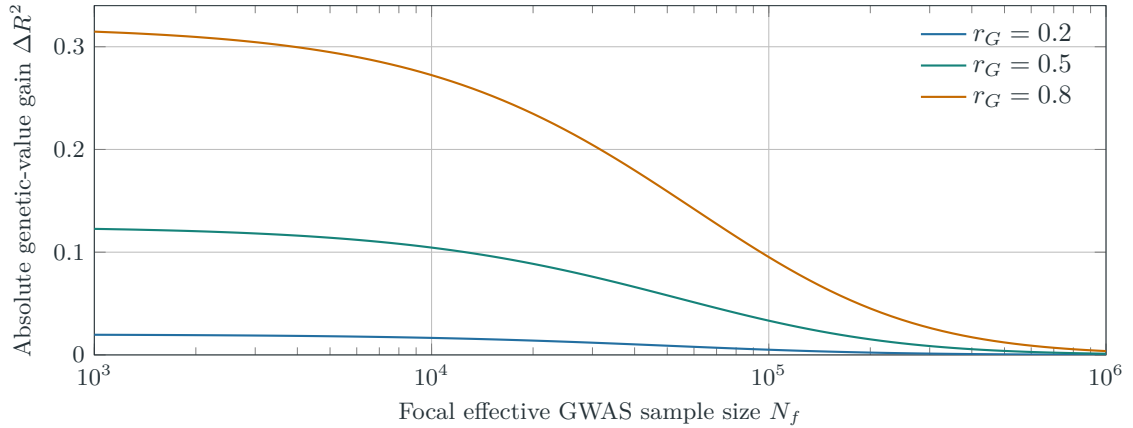


Figure 1: Analytical two-score gain from Equation (6). The illustration fixes $h_f^2 = 0.2$, $M = 20,000$, and auxiliary $R_k^2 = 0.5$, so $x_f = N_f/100,000$. It is an equation plot, not an empirical performance claim.

3 Implemented methods

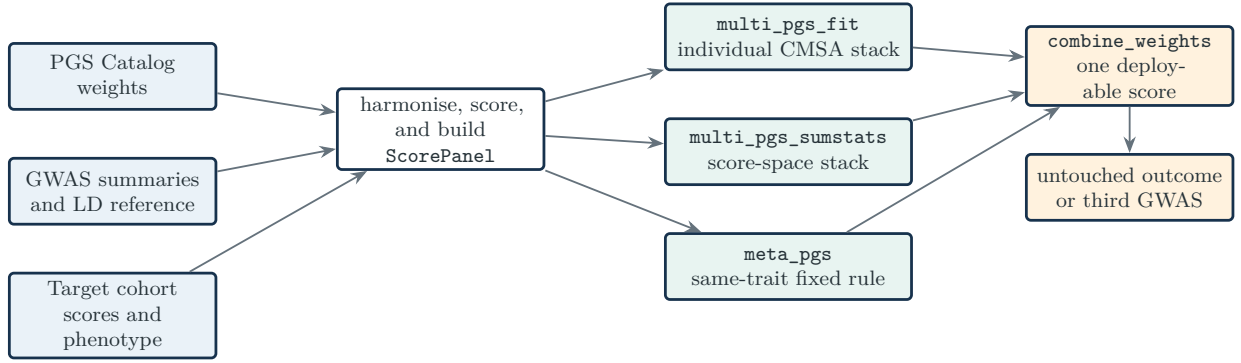


Figure 2: End-to-end package workflow. The three combination routes solve different information problems and are not interchangeable. Assessment data must be untouched by component discovery, combination fitting, and tuning.

Table 3: Information and validation contract for the three fitting routes.

Route	Learns from	Selection / tuning	Valid assessment
multi_pgs_fit	Individual score matrix, target phenotype, optional covariates	Inner CMSA choices inside outer folds; final coefficients average fold-selected fits; nested heuristic may return the baseline	Independent individuals who trained neither component scores nor the combination
multi_pgs_sumstats	Target-GWAS cross-moments plus external LD score Gram	Same moments, an independent second GWAS, or PUMAS-style pseudo-splits	A third untouched GWAS or independent individual-level cohort; tuning performance is not assessment
meta_pgs	Same-trait scores plus effective sample sizes or expected accuracies; target score correlations for decorrelation	No fitted tuning; ridge only stabilises inversion	Independent target outcomes; simple metadata-derived weights remain hypotheses until tested

3.1 Individual-level penalised stacking

Within each training fold, score columns are imputed and standardised. With unpenalised covariates X , the Gaussian or binomial elastic-net objective is

$$\min_{b_0, \gamma, \theta} \frac{1}{2n} \mathcal{L}(y, b_0 + X\gamma + Z\theta) + \lambda \sum_{k=1}^q p_k \left\{ \alpha |\theta_k| + \frac{1-\alpha}{2} \theta_k^2 \right\}. \quad (8)$$

Covariates receive penalty factor zero. The Gaussian path uses covariance updates; the binomial path uses IRLS with coordinate descent, following the glmnet family of algorithms [4]. For a standardised Gaussian column with current partial score z_k^* , the coordinate update is

$$\theta_k \leftarrow \frac{\text{sign}(z_k^*) (|z_k^*| - \lambda \alpha p_k)_+}{G_{kk} + \lambda(1-\alpha)p_k}. \quad (9)$$

CMSA fits a path inside folds, selects a fold-specific model, and averages the selected coefficient vectors [6]. multipgs adds a separate nested outer-fold decision rule that falls back to the unpenalised baseline when the apparent increment is insufficient. The null benchmark shows that this is a pragmatic guard, not a calibrated hypothesis test: pure noise passed in 7 of 30 runs at $n = 1000$ and 4 of 30 at $n = 5000$.

The averaged support is the union of fold supports. Consequently `selected()` reports predictive weights, not causal traits, unique genetic contributors, or stable feature-selection discoveries. In the checked simulation, support recall is high (0.84–1.00 by regime) while precision is only 0.09–0.20.

3.2 Summary-statistic score-space stacking

Let W_{LD} contain component weights on the LD source’s standardised-genotype scale, D be its LD correlation matrix, and W_{GWAS} contain the same component scores on the GWAS source’s standardised-genotype scale.

The sufficient moments are

$$G_{\text{raw}} = W_{\text{LD}}^T D W_{\text{LD}}, \quad c_{\text{raw}} = W_{\text{GWAS}}^T z. \quad (10)$$

Writing $A = \text{diag}(\sqrt{\text{diag}(G_{\text{raw}})})$, the fitted coordinates use

$$G = A^{-1} G_{\text{raw}} A^{-1}, \quad c = A^{-1} c_{\text{raw}}. \quad (11)$$

This is the same Gaussian objective as Equation (8), but in the span of the supplied scores. At $\alpha = 1$, the lasso selects entire component scores; it is inspired by the SNP-level lassosum quadratic [5], but is not SNP-level lassosum.

For a fixed coefficient vector, summary tuning evaluates

$$\text{MSE}(\theta) = \text{var}(y) - 2\theta^T c + \theta^T G \theta, \quad \tilde{R}^2(\theta) = \frac{(\theta^T c)^2}{\text{var}(y) \theta^T G \theta}. \quad (12)$$

MSE, not squared correlation, selects the path because R^2 is blind to a reversed predictor. An independent second GWAS can tune the path; a third is still needed for assessment. PUMAS-style pseudo-splitting uses a joint-Gaussian/CLT covariance approximation [7]. In the committed calibration, relative covariance error is 3.77% for Gaussian and 10.99% for binary simulations.

3.3 Training-free same-trait combination

The simple rules standardise component scores and use

$$w_k \propto \sqrt{N_{\text{eff},k}} \quad \text{or} \quad w_k \propto \sqrt{\hat{R}_k^2}. \quad (13)$$

The first is a power-limited approximation sourced from pipeline code, not from the Hansen et al. preprint. Under independent estimation errors, the exact same-trait GLS direction is

$$w_k \propto \frac{R_k}{1 - R_k^2}, \quad (14)$$

so direct accuracy weighting is accurate only as every R_k approaches zero. The decorrelated rule is

$$w_\delta \propto (C + \delta I)^{-1} \rho. \quad (15)$$

Its algebra is exact under Equation (5), but small error in ρ is amplified along low-eigenvalue directions of C . Ridge reduces that amplification only by moving the result back toward an undecorrelated weighted sum. If only Daetwyler proxies are available, the checked real CAD diagnostic supports `expected_r2`, not `decorrelated`.

3.4 Panel construction, deployment, and evaluation

`panel_from_catalog` harmonises alleles and constructs scores in one genotype pass. `panel_from_sumstats` fits component scores through ldpred3. Identifier order, genome build, effect orientation, and source-specific dosage standard deviation are part of the statistical contract, not file-format trivia.

Combining component variant weights w_{jk} with raw-score coefficients β_k produces the deployable effect

$$\tilde{w}_j = \sum_{k=1}^q w_{jk} \beta_k. \quad (16)$$

The frozen scoring scale should be carried to new cohorts. Re-standardising each cohort independently can change coefficients, calibration, and even the meaning of a fixed threshold.

With covariates, `multipgs` reports the unnormalised increment

$$R_{\text{incremental}}^2 = R_{\text{covariates+score}}^2 - R_{\text{covariates}}^2. \quad (17)$$

The Albiñana paper instead divides this difference by $1 - R_{\text{covariates}}^2$ [9]; absolute values are therefore not identical even when relative comparisons share covariates.

For binary outcomes, with population prevalence π , sample case fraction p_s , threshold $t = \Phi^{-1}(1 - \pi)$, density $\varphi(t)$, and $i = \varphi(t)/\pi$, the implemented Lee transformation is

$$C_{\text{LT}} = \frac{[\pi(1 - \pi)]^2}{\varphi(t)^2 p_s (1 - p_s)}, \quad \vartheta = \frac{i(p_s - \pi)}{1 - \pi} \left\{ \frac{i(p_s - \pi)}{1 - \pi} - t \right\}, \quad R_L^2 = \frac{C_{\text{LT}} R_O^2}{1 + C_{\text{LT}} \vartheta R_O^2}. \quad (18)$$

This metric depends on a stated prevalence assumption [8].

4 Empirical evidence in the repository

4.1 Individual fitting: estimator sanity

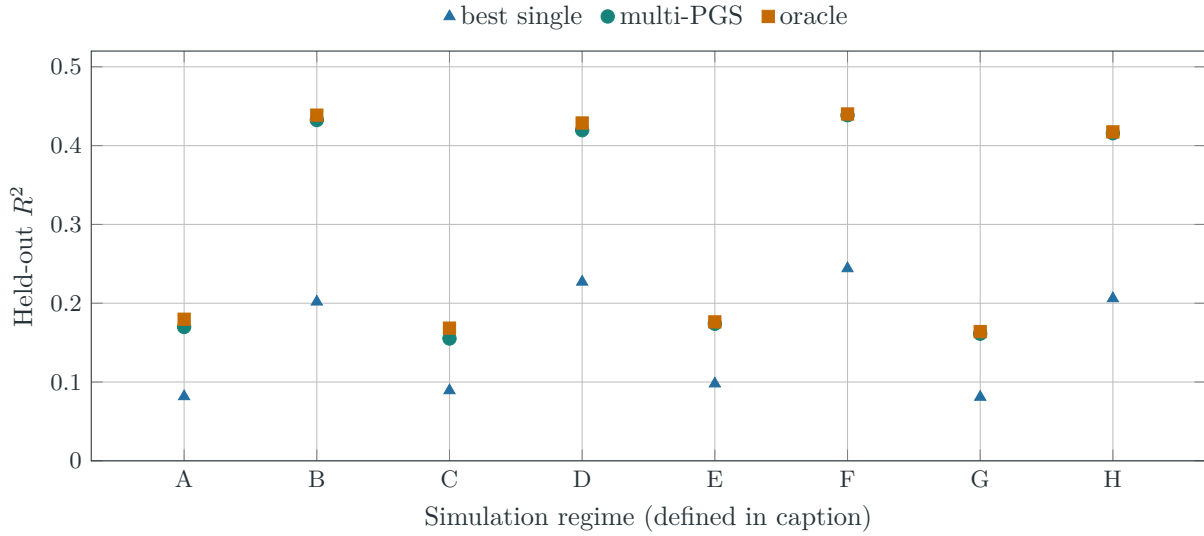


Figure 3: Mean held-out R^2 over 30 seeds from `benchmarks/results/fit_accuracy_summary.csv`. Regimes A–D use $n = 2000$, E–H use $n = 10,000$; A,B,E,F use $q = 50$, the others $q = 200$; A,C,E,G use $h^2 = 0.2$, the others $h^2 = 0.5$. The multi-PGS beats the training-selected best single score in all 240 raw replicates. Because the simulated genetic value lies in the score span, this is an estimator sanity check, not evidence of clinical efficacy.

Across the eight regimes, mean uplift over the selected single score ranges from 0.0660 to 0.2307 R^2 . Based on regime means, the fitted stack closes 83.4%–99.2% of the gap between the selected single score and the oracle. The overall mean nested-CV minus held-out difference is -0.00230 , slightly conservative in this simulation.

4.2 Summary moments and LD-reference sensitivity

With exact individual moments, the summary and individual standardised coefficients correlate 0.999624 on average, and all compared held-out R^2 values lie between 0.4973 and 0.4982. This supports the score-space identity under the benchmark’s strong-signal, small- q design.

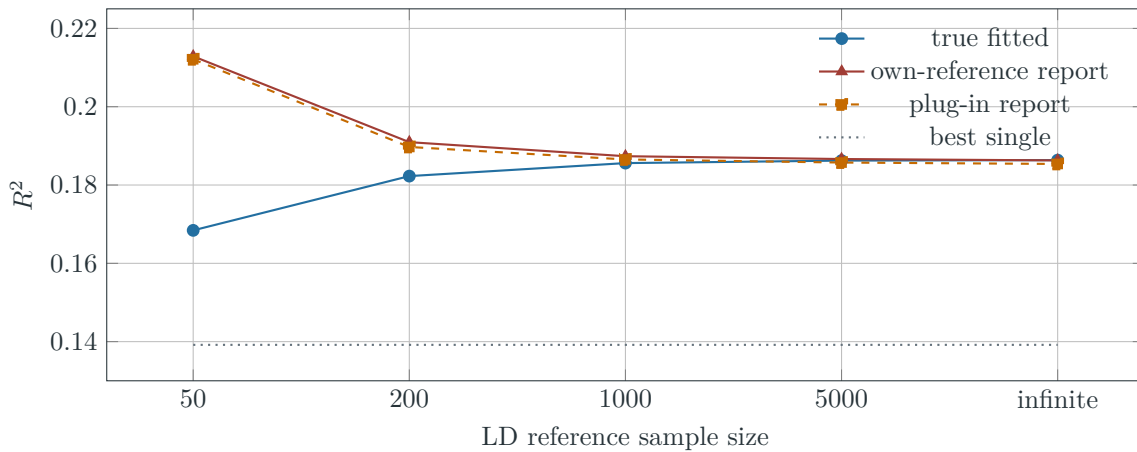


Figure 4: Real-LD simulation from `real_ld_simulation_summary.csv`. At reference $n = 50$, true fitted R^2 falls by 0.01795 relative to the infinite-reference result, while the own-reference estimate is inflated by 0.04446 (26.4% of true fitted R^2). Reporting bias is larger than the loss in the fitted predictor.

4.3 Overlap and decorrelation

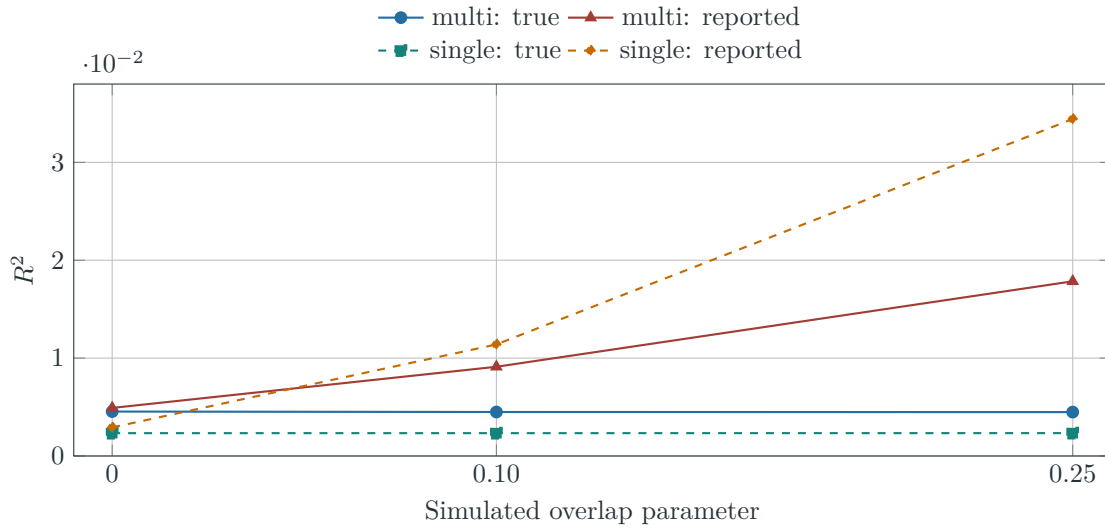


Figure 5: Known-overlap simulation from `overlap_inflation_summary.csv`. At overlap 0.10, reported-to-true R^2 ratios are 2.03 for multi-PGS and 4.89 for the single score; at 0.25, they are 3.98 and 14.73. Bivariate LDSC detected all 16 non-null runs and none of eight null runs in this 109,590-variant design. This does not imply universal LDSC power, but it refutes the absolute claim that no downstream diagnostic can ever flag overlap.

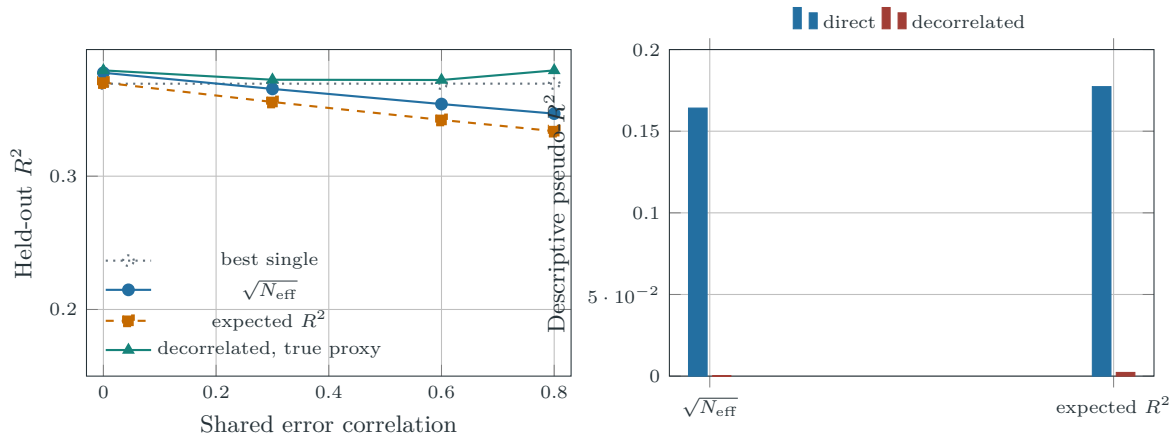


Figure 6: Decorrelation behaves well when the simulation supplies the intended accuracy structure (left, 30 seeds per point), but collapses with proxy accuracies in the single checked 24-score CAD diagnostic (right). The right panel is a contaminated regime-C score-space description, not external outcome truth. Sources: `meta_rules_summary.csv` and `real_meta_rules_summary.csv`.

Table 4: Headline checked evidence, separated by evidence level.

Artifact	Result	What it supports - and does not
fit_accuracy	Multi-PGS wins 240/240 held-out simulations; uplift 0.0660–0.2307	Supports estimator sanity when truth lies in the score span; does not establish real-outcome or clinical superiority.
null_gate	Noise passes 11/60 runs overall	Shows the fallback is a decision heuristic, not a significance test.
sumstat_vs_individual	coefficient correlation 0.999624; R^2 0.4973–0.4982	Supports exact-moment Gaussian equivalence in a small- q , strong-signal design.
sumstat_calibration	PUMAS covariance relative error 3.77% Gaussian, 10.99% binary	Quantifies approximation error; it does not validate pseudotuning as assessment.
real_ld_simulation	reference $n = 50$ report inflation 0.04446	Demonstrates reference-induced optimism under real LD.
overlap_inflation	up to 3.98-fold multi-PGS and 14.73-fold single-score inflation	Demonstrates overlap danger under a known covariance construction.
real_meta_rules	direct expected- R^2 proxy 0.17729 versus decorrelated 0.002261	Shows severe proxy sensitivity in one regime-C CAD diagnostic; cannot rank methods externally.

5 Technical review findings

5.1 Confirmed independent-tuning defect

The helper `_range_basis_from_factor` normalises the columns of an arbitrary covariance factor independently. Column normalisation does not make an arbitrary factor orthogonal. Independent tuning then rescales this factor by training score standard deviations and applies BB^T as a projection. The operation is a projection only if

$$B^T B = I \implies P = BB^T, P^T = P, P^2 = P. \quad (19)$$

The implication’s premise generally fails.

A public two-score reproduction uses the full-rank tuning Gram

$$G_v = \begin{pmatrix} 4 & 3 \\ 3 & 9 \end{pmatrix}. \quad (20)$$

Because G_v is full rank, its range is all of $\mathbb{R}^2$, so the correct range projection is the identity. With unequal training and tuning score scales, the implementation changes training $c = (0.2, 0.1)$ to $(0.391198, 0.004401)$ and logs discarded fractions of 0.95599 for both training and tuning.

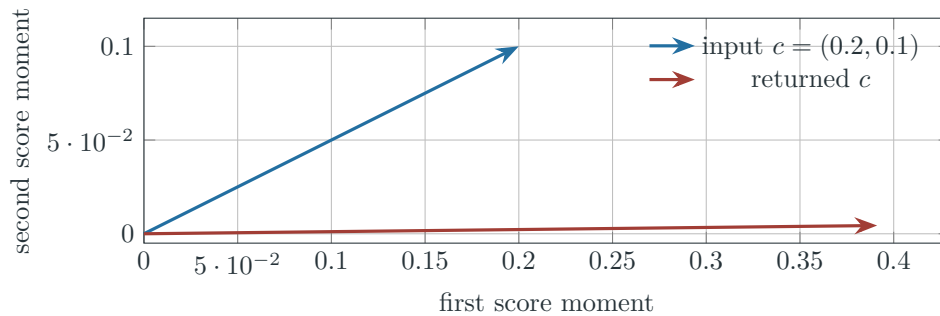


Figure 7: Independent-tuning counterexample. A projection onto the range of a full-rank 2×2 Gram must leave every vector unchanged. The large rotation and rescaling therefore prove a correctness defect rather than a numerical tolerance dispute.

The simplest repair is to derive the independent-tuning basis directly from `_range_basis(sel_gram)`, or to compute a rank-revealing QR/SVD basis for the scaled factor. Regression tests should assert orthonormality, symmetry/idempotence, full-rank identity, scale invariance, and correct rank-deficient behaviour.

Table 5: Prioritised review findings.

Priority	Location	Finding and consequence	Recommended action
High	sumstat.py: 1002-1012, 1326-1338	Non-orthogonal factor columns are used as an orthonormal range basis. Independent tuning can distort all moments even for full-rank tuning LD.	Block the independent route; use an eigendecomposition or QR/SVD basis; add the five regression tests described above.
High	README.md:125-132; guide.md:441-445	The showcased Daetwyler-to-decorrelated recipe contradicts code comments and the checked CAD diagnostic, where R^2 falls 78-fold.	Show <code>expected_r2</code> for metadata proxies. Reserve decorrelation for accurate, correctly oriented component accuracy and require external assessment.
Medium	theory.md:175-190; algorithm.md:5-20	The objective is written on raw scores, while implementation penalises fold-standardised coefficients. Penalty factors are misinterpreted under the raw equation.	Define Z, θ, β as in Equation (1) and write the objective as Equation (8).
Medium	algorithm.md:299-343	Moment-derived in-sample quantities are called <i>true</i> and an untracked oracle table reports R^2 0.542; the committed run is regime C.	Replace with checked artifact values, label pseudo- R^2 , and store any future oracle/ridge sweep in machine-readable results.
Medium	sumstat.py:1970-1998	<code>align_to_reference</code> promises standardised-genotype weights even when no SD conversion is requested. Ignoring a warning can create plausible, incorrect moments.	Require explicit scale or make the raw/standardised return contract explicit in both type and documentation.
Medium	stack.py:471-476; _coord.py: 208-233, 370-420	Several numerical controls are not validated and solvers expose no convergence status. Invalid settings or exhausted iterations can silently return partial paths.	Validate public controls and report convergence/iteration counts per path point.
Medium	README.md:82; stack.py:selected	Selected scores are described as traits that <i>carried the signal</i> . CMSA support is unstable and predictive, not causal.	Describe largest nonzero standardised weights; prohibit causal or genetic correlation interpretation.
Medium	benchmarks/README.md; provenance JSON	Several prose values are stale; three real-data files record ldpred3 0.3.1, outside the current dependency range; no git or script hash is recorded.	Generate prose tables from CSV, record git and script hashes, and regenerate or clearly grandfather incompatible runs.
Low	meta.py:324-332	<code>expected_r2</code> accepts finite values above one despite its documented R^2 meaning.	Reject values outside $[0, 1]$, or rename the argument as an arbitrary positive proxy.

6 Documentation and reproducibility assessment

The documentation’s strongest feature is its willingness to state operational failure modes: target/discovery overlap, relatedness, ancestry mismatch, genome-build alignment, tuning-versus-assessment separation, and liability prevalence. The bibliography is extensive and documentation tests require cited DOIs to appear in the

annotated reference list.

The main weakness is traceability. Numerical claims are copied manually into prose, some diagnostic values exist only in stdout, and benchmark provenance does not identify the exact source commit or script hash. As a result, a scientifically careful paragraph can still be numerically stale.

Table 6: Reproducibility and validation matrix.

Area	Current strength	Material gap
Tests	294 passing tests cover API contracts, alignment, fitting, numerical identities, and failure cases.	No coverage metric; no benchmark execution in CI; the independent-tuning bug lacks a full-rank unequal-scale test.
Benchmark records	Eleven producer scripts and 34 artifacts include raw seed rows, summaries, commands, timestamps, versions, and many input hashes.	Some prose numbers drift; several real runs use incompatible ldpred3 metadata; git commit and script hash are absent.
Scientific scope	Simulations test null behaviour, exact moments, overlap, real LD, and score-correlation rules.	No real individual-level outcome benchmark, ancestry transfer, replicated binary accuracy study, or competing stacker comparison.
Documentation tests	Internal paths, benchmark names, bibliography shape, and DOI membership are checked.	Anchors, snippets, factual citation metadata, and CSV-to-prose numerical consistency are not checked.
Distribution	README, Markdown docs, examples, tests, and benchmarks are included in the source distribution.	Before this report, LaTeX/PDF artifacts were absent from both source and wheel distributions; broad benchmark grafting can also capture ignored local data in developer builds.

7 Interpretation limits

- No causal interpretation.** Coefficients optimise prediction under correlation and penalisation. They do not estimate genetic correlations, mediation, or trait-specific causal contributions.
- No automatic overlap cure.** Cross-validation inside a target cohort cannot remove discovery contamination shared by every fold. Bivariate LDSC or named-cohort metadata may flag overlap under favourable conditions, but exclusion by design is stronger.
- No ancestry model.** Fitting weights in a target cohort does not repair discovery-score or LD-reference non-portability. Absolute risk and score means are particularly unsafe across ancestries.
- Direction matters.** Squared correlation hides sign reversal, and the training-free expected- R^2 routes assume consistent positive trait orientation. Report signed correlation or calibration slope alongside R^2 .
- No clinical calibration.** A high ranking metric is not a calibrated probability. Calibration intercept/slope, subgroup performance, decision utility, and an externally justified prevalence are separate requirements.
- Conditional uncertainty.** The package bootstrap conditions on fixed component scores and their discovered weights; it does not propagate component-GWAS, LD-reference, or architecture-estimation uncertainty.

8 Recommended roadmap

Table 7: Recommended sequence of work after this review.

Order	Work item	Acceptance criterion
1	Fix independent tuning	Full-rank unequal-scale moments are unchanged; projection is orthogonal and idempotent; scale-invariance and rank-deficient tests pass.
2	Correct high-impact guidance	README and guide no longer recommend decorrelation from Daetwyler proxies; scale and support interpretations match code.
3	Make evidence self-checking	Report tables/figures and key docs are generated or tested against CSV; every benchmark records git commit, script hash, exact dependency revision, and external-input identities.
4	Expose convergence	Public numerical controls are validated and every solver result reports convergence and iteration exhaustion.
5	Add outcome validation	Replicated real individual-level continuous and binary outcomes, untouched assessment, ancestry strata, calibration, and at least one comparator stacker.

9 Conclusion

`multipgs` has a coherent central idea: move a difficult multi-trait genomic problem into a smaller space of component scores, then choose weights using the information actually available. The selection-index derivation explains when gains are possible; the individual and summary Gaussian objectives explain how weights are learned; and the same-trait rules provide a transparent fallback when phenotypes are unavailable.

The package is not ready to treat every route as equally mature. Individual fitting has strong controlled simulation support but needs external outcomes. Summary fitting has convincing moment checks but its independent-tuning route has a confirmed correctness defect. Training-free decorrelation is an oracle formula whose practical success depends on an accuracy vector the package usually cannot know. Fixing that distinction in both code and prose is the highest-value next step.

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A Artifact provenance

Table 8: Files used directly for quantitative statements and figures.

Repository artifact	Use in this report
benchmarks/results/fit_accuracy.csv and fit_accuracy_summary.csv	Individual-fit held-out accuracy, uncertainty, support, and CV calibration.
benchmarks/results/null_gate.csv and null_gate_summary.csv	Pure-noise and signal fallback behaviour.
benchmarks/results/sumstat_vs_individual_summary.csv	Individual/summary coefficient and held-out accuracy agreement.
benchmarks/results/sumstat_calibration_summary.csv	Exact-moment identity and PUMAS covariance approximation error.
benchmarks/results/real_ld_simulation_summary.csv	Real-LD reference-size sensitivity and reporting bias.
benchmarks/results/overlap_inflation_summary.csv	Known-overlap accuracy inflation and LDSC diagnostic rate.
benchmarks/results/meta_rules_summary.csv	Synthetic same-trait rule sensitivity over shared residual correlation.
benchmarks/results/real_meta_rules_summary.csv	Single real CAD regime-C descriptive comparison.

Three real-data provenance records report ldpred3 0.3.1, whereas the reviewed `pyproject.toml` requires ldpred3 0.4.5–0.4.x. These records may reflect stale editable-install metadata or an unsupported runtime; either way, exact environmental provenance is unresolved. This report therefore uses them as diagnostics with explicit caveats, not as current-version validation.

B Independent-tuning reproduction

The defect in Figure 7 is reproduced with public API inputs:

$$W_{\text{train}} = \text{diag}(1, 10), \quad z_{\text{train}} = (0.2, 0.01)^T, \quad W_v = \text{chol}(G_v)^T, \quad z_v = (W_v^T)^{-1}(0.4, 0.7)^T. \quad (21)$$

Calling `multi_pgs_sumstats` with identity LD for both sources, `tune="independent"`, and `ld_shrinkage=0.1` returns the altered training moment shown in Figure 7. The correct full-rank projection is identity regardless of the shrinkage used later in the fitted quadratic.